

Arman Hossen

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Portfolio · LinkedIn · Google Scholar

Statistician and researcher turning complex health data into actionable insights. Focused on non-communicable disease, maternal health, and mental health outcomes through predictive modeling, machine learning, and large-scale national survey data.

Education

Master of Science (MS) in Statistics **Fall 2025 – Present**
Miami University, Oxford, Ohio, USA *CGPA: 3.65/4.00*

- Relevant Coursework: STA 502 Statistical Programming (SAS), STA 504 Advanced Data Visualization (R), STA 563 Regression Analysis, STA 566 Experimental Design Methods, STA 664–665 Theory of Statistics I & II

Master of Science (MS) in Statistics **January 2023 – March 2024**
Dhaka College (affiliated with the University of Dhaka), Bangladesh *CGPA: 3.86/4.00*

- Graduated 1st in class

Bachelor of Science (BS) in Statistics (Honours) **January 2018 – December 2022**
Dhaka College (affiliated with the University of Dhaka), Bangladesh *CGPA: 3.84/4.00*

- Graduated 1st in class

Research Interests

Biostatistics • Predictive Modeling • Machine Learning • Neural Networks • High-Dimensional Biomedical Data • Model Evaluation & Calibration • Statistical Inference • Reproducible Research Pipelines

Professional Experience

Junior Officer, Data Analytics **January 2025 – May 2025**
Development Research Initiative, Bangladesh

- Developed 100+ digital survey forms in SurveyCTO and KoboToolbox for research projects led by institutions including the World Bank, Oxford, LSE, Purdue, UPenn, and BRAC.
- Developed 10+ live dashboards in Google Sheets and Power BI, integrated with SurveyCTO API for real-time data monitoring.
- Provided feedback on question design and structure; conducted training sessions for field researchers.

Research Assistant Intern **November 2024 – December 2024**
BRAC James P. Grant School of Public Health, Bangladesh

- Assisted four international MPH students in designing and developing research methodology.
- Developed comprehensive questionnaires for Key Informant Interviews (KII) and survey data collection.
- Managed and analyzed data using Microsoft Excel and Stata.

Teaching Experience

Graduate Assistant (Teaching) **Fall 2025 – Present**
Department of Statistics, Miami University, Oxford, Ohio, USA

- Serving as Graduate Assistant for STA 261 (Statistics), supporting course delivery and undergraduate instruction within the Department of Statistics.
- Instruct weekly lab classes using JMP statistical software, guiding students through hands-on data analysis exercises.
- Hold office hours and provide academic support to 115 undergraduate students every semester.

Tutor – Statistics and Mathematics **2022 – 2025**
• Tutored over 200 undergraduate students across: Introduction to Statistics, Introduction to Probability & Probability Distributions, Linear Algebra, Calculus I & II, and Ordinary Differential Equations.

Publications & Research

Peer-Reviewed Journal Articles

1. Nahin, K. S. A., & Hossen, A. “Prehypertension in Bangladesh: Evidence from BDHS 2022.” *Frontiers in Public Health* (Published, March 2026). DOI: 10.3389/fpubh.2026.1777932

- Rony, A. R., Hossen, A., Asha, A. S., & Nahin, K. S. A. "From Medical Necessity to Private-Sector Dominance: A Two-Decade Evolution of Caesarean Section Determinants in Bangladesh (2004–2022)." *Scientific Reports* (Published, May 2026). DOI: 10.1038/s41598-026-52229-9
- Hossen, A., et al. "Reproductive Coercion as a Determinant of Anxiety and Depression Among Married Women in Bangladesh: A Nationally Representative Survey-Weighted Multinomial Analysis of the 2022 DHS." *Archives of Women's Mental Health* (Under Review, submitted January 2026).
- Hossen, A., et al. "Socioeconomic Inequality and Postpartum Depression in Bangladesh: Evidence from BDHS 2022." *Humanities and Social Sciences Communications* (Under Review, submitted March 2026).
- Hossen, A., et al. "The Paternal Shift in Child Survival: A Zero-Inflated Modelling of Socioeconomic Influence and Parental Education on Under Five Children Death Per Mother in Bangladesh using BDHS 2022." *Scientific Reports* (Under Review, submitted February 2026).
- Nahin, K. S. A., Hossen, A., & Jannatul, T. "Shifting Prevalence and Risk Factors of Non-Communicable Diseases in Bangladesh: A Comparative Multilevel Analysis of Nationally Representative BDHS Data (2017–2022)." *medRxiv* (Preprint, posted April 2026). DOI: 10.64898/2026.03.31.26349897

Conference Abstracts & Presentations

- Ahmed, S. M. F., Asha, A. S., Nahin, K. S. A., & Hossen, A. "Neighborhood Influences on Mental Health in Older Age: Evidence from the 2020 Health and Retirement Study (HRS)." Gerontological Society of America (GSA) Annual Scientific Meeting 2026, National Harbor, MD. (Poster Presentation, Abstract ID: 2376958).

Undergraduate Research Project

"A Comparative Study of Academic Performances of Residential and Non-Residential Students of 7 Colleges Affiliated with the University of Dhaka."

Supervisor: Associate Professor Runa Laila, Department of Statistics, Dhaka College

- Designed and administered a questionnaire for primary data collection.
- Organized data and conducted univariate and bivariate statistical analyses.

Technical Strengths

Statistical Software	RStudio, SAS, JMP, IBM SPSS, Python, Stata
Data Collection Tools	SurveyCTO, KoboToolbox
Office Tools	MS Word, MS Excel, MS PowerPoint

Standardized Test

IELTS Overall Band Score: 7.0 — Listening: 7.5 Reading: 8.0 Writing: 6.5 Speaking: 6.5

Professional Certifications

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- Information and Communication Technology (ICT) – Dhaka College, University of Dhaka — GPA: 4.00/4.00 (2022–23)
 - DataCamp Courses (Completed):** Introduction to R; Introduction to the Tidyverse; Introduction to Data Visualization with ggplot2
 - DataCamp Courses (In Progress):** Data Visualization in Tableau; Building Dashboards with shinydashboard

Achievements

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- Regional Round Champion and National Round Participant – 7th Bangladesh Physics Olympiad (2017)
 - District Round Champion – Mathematics Olympiad, Lakshmipur (2016)

Awards and Leadership

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- Organizational Chair, Bangladesh Student Association (BSA), Miami University** (2026 – Present). Coordinate member engagement and community programs.
 - Vice President – Data Analytics Club, Dhaka College (DACDC)** (July 2022 – June 2023). Led club initiatives, workshops, and seminars.
 - District Volunteer Coordinator – UNICEF Bangladesh** (June 2021 – January 2023). Supervised volunteer teams and supply distribution across the district.
 - Campus Ambassador – Youth in Digital Awareness (YDA)** (January 2018 – July 2019). Organized digital-awareness workshops and campaigns.
 - Executive Member – Ex-Farukyans' Association** (2018 – 2025). Organized alumni reunions and networking events.